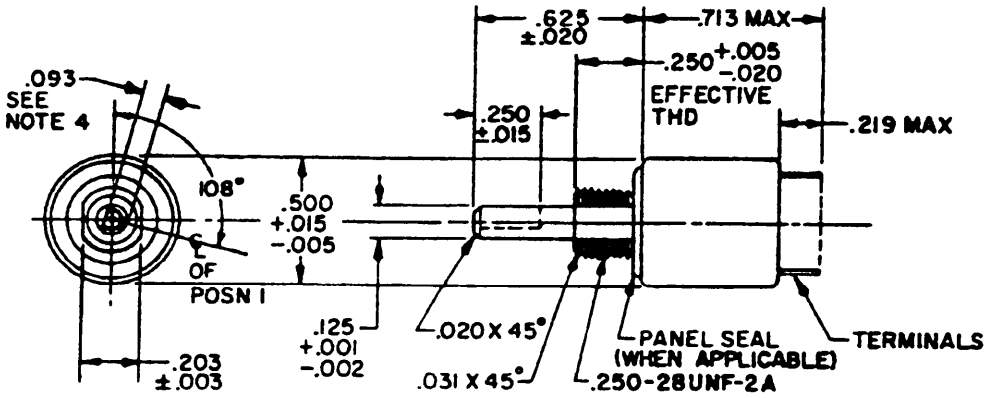
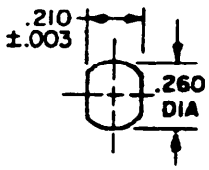
MILITARY SPECIFICATION SHEET

SWITCH, ROTARY, CLOSED CONSTRUCTION, EXPLOSION PROOF, 1/2 INCH DIAMETER, CODED OUTPUT CIRCUITRY STYLE SR37

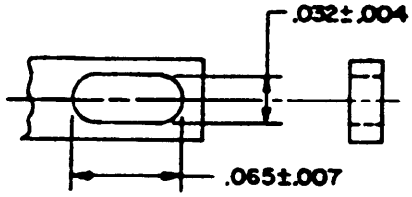
Ⓐ

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Specification MIL-S-3786.

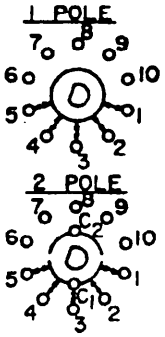




MOUNTING HOLE



SOLDER LUG TERMINAL DETAIL



CIRCUIT DIAGRAM
(SWITCH IS VIEWED FROM SHAFT END AND SHOWN IN POSITION 1)

INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.020	.51	.210	5.33
.002	.05	.031	.79	.219	5.56
.003	.08	.032	.81	.250	6.35
.004	.10	.065	1.65	.260	6.60
.005	.13	.093	2.36	.500	12.70
.007	.18	.125	3.18	.625	15.88
.015	.38	.203	5.16	.713	18.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .005 (.13 mm) and ± 3° on angles (non-accumulative).
4. Shaft flat is located diametrically opposite to the number one terminal. Position 1 and terminal 1 coincide

FIGURE 1. Coded output switch.

Ⓐ denotes changes

FSC 5930

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REQUIREMENTS:

Dimensions and configuration: See figure.

Angle of throw: 36°.

Construction style: See table III.

Insulation: Symbol P (plastic).

Mounting hardware: Each switch shall be provided with one corrosion-resistant steel hexagon nut, .089 ± .010-inch thick by .375 inch across the flats and one internal-toothed lock washer, .396/.408 diameter.

Moisture resistance: Insulation resistance, measured immediately after conclusion of the moisture-resistance test and while the switches are still in the humidity chamber, shall be greater than 100 megohms. At the end of the drying period, the insulation resistance shall be not less than 5,000 megohms.

Ⓐ Temperature-life characteristic: 10,000 cycles (-65° C and +85° C).

Vibration grade: 3 (10 to 2,000 Hz).

Shock symbol: H (high impact), and
M (medium impact).

Altitude: C (70,000 ft).

Terminal strength (pull): A force of 1.5 pounds shall be applied to the terminals.

Terminal marking: Switches may be marked with terminal numbers on side of switch.

Ⓐ Switching characteristic: Nonshorting.

Rotation: See table III.

Rotational torque: The minimum and maximum values of torque determined for shaft rotation shall be within the limits specified in table I.

TABLE I. Rotational-torque limits.

Temperature	Torque (lb-in)	
	Minimum	Maximum
Room	.4	1.5
Minimum	.4	2.0

Life (rotational): One switch shall be tested to the load specified for the applicable environmental condition. All rotor contacts (5) of each switch shall switch the applicable load.

Ⓐ Extent of qualification: "F" construction style qualifies "E" construction style for the same switches provided two sample units of "E" construction switches are tested to explosion requirement paragraph to verify explosion proof capability.

Qualification may be extended to products produced by manufacturers on QPL-3786 20, provided two sample units are subjected to and pass the 10,000-cycle endurance test, and the rotational torque, contact resistance, and dielectric withstanding voltage tests both prior to and following the endurance test.

The activity responsible for this specification is Air Force, Code 11; however, qualification will be administered by DESC-EQ.

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Ⓐ Electrical loads: 20 milliamperes maximum resistive at 28 Vdc maximum at both atmospheric and reduced barometric pressure.

Ⓐ TABLE II. Truth table indicating circuitry.

Switch position	Count	X denotes circuit is made: Switch terminal 1	2	3	4	5
1	0	X	X	X	X	X
2	1		X	X	X	X
3	2			X	X	X
4	3				X	X
5	4					X
6	5					
7	6	X				
8	7	X	X			
9	8	X	X	X		
10	9	X	X	X	X	

Ⓐ Dielectric withstanding voltage: The magnitude of test voltage for the dielectric withstanding voltage shall be 50 volts, alternating current (ac).

Ordering data: Procurement documents should specify the following:

(a) Title, number, and date of this specification sheet, and the military part number as listed in table III.

Ⓐ TABLE III. Switching characteristics and part numbers.

Part No. M3786/37- Construction: F	Part No. M3786/37- Construction: E	Number of poles	Pos. per pole
001	301	1	10 1/
002	302	2	5 1/
003	303	1	10
004	304	2	5

Ⓐ Application note: Switches covered by this document are intended only for special low-current, low voltage applications requiring coded circuitry. For general service requirements of similar switches with conventional circuitry, refer to MIL-S-3786/20.

1/ Continuous rotation.

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5930-F401)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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